

PLS-SEM-Based Evaluation of Composite Material Integration in Aerospace and Automotive Engineering

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Abstract— In the present study, we examined the key determinants influencing the adoption of composite materials in crucial industries such as the automotive and aerospace domains. A structured online survey was distributed to professionals in the industry (N=74), including engineers, managers, and technical specialists, to obtain empirical data for the model. Data analysis was conducted utilizing the Partial Least Squares Structural Equation Modelling (PLS-SEM) technique via SmartPLS 4.0. Three key parameters were analyzed; design and manufacturing, cost-effectiveness, and durability and utilization. Using PLS-SEM enabled the analysis of both direct and indirect relationships between the parameters, allowing the adoption of composites to be analyzed holistically rather than independently. We found that design and manufacturing quality have a direct impact on cost-effectiveness ($R^2 = 0.538$) and long-term durability ($R^2 = 0.532$), emphasizing the interconnected nature of these factors as well as the dual and compound influence of design and manufacturing on the adoption of composites. The internal reliability of the study was validated via parameters such as Cronbach's α (0.816–0.877) and composite reliability (0.867–0.911), which confirmed that the measurements were consistent across the three latent constructs. The findings have some significant implications for how firms approach the implementation of composite materials. The study fills a research gap by holistically modelling the latent factors rather than assessing them independently. This provides a robust framework for application. These insights are particularly relevant in the aerospace and automobile industries, where properties such as lightweighting and durability are highly valued.

Keywords— Composite materials; Aerospace engineering; Automotive manufacturing; PLS-SEM; Cost-effectiveness; Durability

I. INTRODUCTION

Ever since their inception, composite materials (CMs) have repeatedly demonstrated their potential and efficacy as crucial elements of most engineering marvels today. They are mixtures of two or more materials, which, when combined, put forth the best characteristics of each material or manifest an entirely unique set of characteristics that no single material can possess on its own [1][2]. Their widespread use in modern

times is due to their adaptability and ability to serve in a variety of situations and applications. These features and characteristics are highly valued in the automotive and aerospace industries. Another crucial feature that is highly valued in these industries is performance, which CMs bring to the table through a variety of fundamental characteristics such as superior strength-weight ratio, mechanical properties, and thermal properties [1][2].

The high strength-weight ratio is particularly crucial as this will lead to weight savings, in turn impacting all-round efficiency and cost savings [3][4]. The data confirms the exponentially growing demand for composites, with the global polymer market surpassing USD 100 billion in 2022. Moreover, the aerospace and automotive industries combined made up nearly 50% of the market shares [3][4].

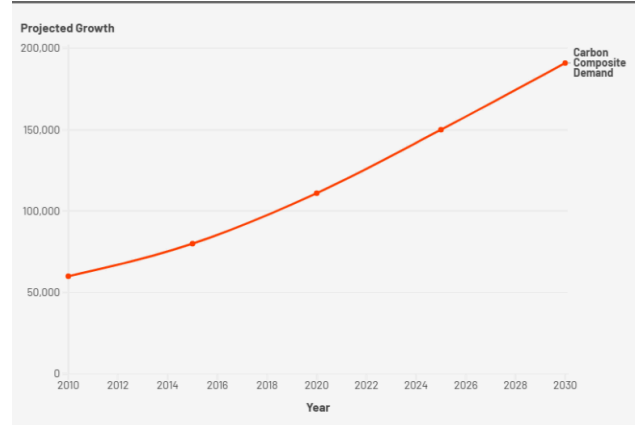


Figure 1: Projected demand for Carbon Composites

The continued growth in demand for composites and the consequent growth of the market, specifically for carbon composites, is demonstrated in fig. High-Performance Advanced Composites (HPAC), particularly fibre-based composites, have considerable applications in the aerospace and automobile fields by leveraging materials such as carbon, glass, aramid, and nanofibers. Composites, however, are used widely in the commercial sector as well. After transitioning from light

alloys to composites, the Airbus A320 achieved an 800kg weight reduction by using them in both primary and secondary structures, saving 400kg of aluminium alloys. The Boeing 787 Dreamliner profited from a similar transition; it benefited from nearly a 20% weight reduction and the elimination of a thousand mechanical fasteners per fuselage barrel by implementing structures composed of nearly 50% Carbon Fibre Reinforced Polymers (CFRPs). More impressively, the structural integrity of composites, with some unidirectional carbon composites being 3x stronger than aluminium alloys, ensured that safety was not compromised in the hunt for fuel efficiency and cost savings. These benefits are visually represented in fig. 2.

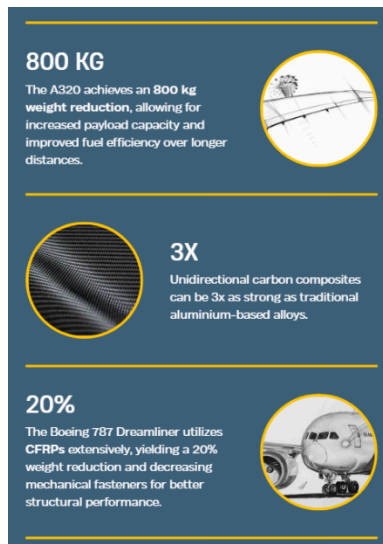


Fig. 2: The quantitative benefits of composites

The shift towards composite materials is driven mainly by considerable performance gains and economic benefits. Environmental impacts are also substantial, as weight reductions can result in fuel savings and reduced CO₂ emissions; every kilogram of weight saved on commercial airliners is said to save approximately 2900 liters of fuel annually [5]. Specific sectors value CMs for different reasons. The aerospace industry, for example, values composites due to properties such as high stiffness, low weight, and corrosion resistance [6]. This extends to deep space applications as well, where weight reductions between 25-30% are projected to lead to compounded cost savings of nearly 50% [6]. A similar, albeit less exponential, growth in the applications of CMs in the automobile industry has occurred. The structures of newer vehicle models are made up of nearly 50% composites and polymers, resulting in lower weight, improved fuel efficiency, and greater crash energy absorption [7]. There exists a further scope for innovation in the automobile industry: CMs such as Natural Fibre Composites (NFCs) and wood-based composites are emerging as viable alternatives to the traditional metal alloys used in the construction of vehicles [7].

Considering all the advantages and the commercial viability of composites, it seems imprudent not to incorporate composites on a larger scale. However, the transition to

composites is not without its challenges. In deep space applications, engineers have difficulties in predicting failure points, meaning they often must overdesign structures involving composites. Hence, improvements in analysis and modelling software are necessary [8]. Capabilities regarding materials, processes, automations and general R&D also require development for composite adoption to increase. In terms of mass production, the supply chain presents the biggest issues as they are immature and lack technological infrastructure [9]. Additionally, there is a need for the development of new manufacturing processes like Out-of-Autoclave (OOA) curing, which, due to the size factor and extreme requirements of space, emerges as the largest cost driver for space agencies such as NASA [7]. Deep space applications also require a level of multifunctionality (materials that offer radiation protection, impact resistance and structural integrity) that composites can fulfil in the future but require further development [10]. Moving back to commercial aviation, aluminium-, magnesium-, and titanium-based alloys, along with advanced ceramic and polymer composite systems, have been developed with some exceptional properties. Despite these advancements, persistent limitations remain, including mechanical degradation under load, corrosion susceptibility, stress-induced cracking, and fretting-related surface damage [11]. Lastly, issues such as recyclability, scalability and reliable quality assurance remain key concerns for the adoption of composites [12].

The existing literature documents the implementation of composites into the aerospace and automotive sectors to a large extent, particularly on the ability of composites to enhance performance with minimal increases in weight and maintenance [13]. These papers are centred around the key themes of material properties; they investigate the physical and mechanical behaviour of composites such as CFRPs and polymer matrix composites [13][14]. Other papers study the applications of more innovative composites like bio-inspired self-healing graphene-based composites. These composites imitate the healing process found in nature, thereby improving the reliability of critical components in the aerospace, automobile and electronics industries by implementing autonomous repair mechanisms [15]. These studies use techniques like microscopy to understand the behaviour of said materials at the micro-level [15].

Evidently, there is a substantial amount of research on the capabilities of composites regarding different parameters (properties, innovativeness etc.). However, the research often addresses these parameters in isolation from one another; there exists a gap in studies that model the complex relationship between parameters such as operational performance, cost-effectiveness and the economic factors. This “grey area”, where the relationships between parameters must be measured indirectly, is lacking in existing literature and can be explored using latent variable modelling [16].

Therefore, for this gap to be filled, it is necessary to move

to a broader viewpoint and shift from an individual analysis of parameters to a more holistic approach. This paper looks to address an important research question: What are the major factors and outcomes of the adoption of composite materials in automotive and aerospace domains when modelled as an **interconnected system**? To successfully answer this question, our primary objective is to model said factors and outcomes using PLS-SEM. Being a multivariate analysis technique, PLS-SEM is a strong candidate for this type of research as it allows an empirical and simultaneous analysis of multiple variables and latent variables through measurable and quantifiable indicators [16]. This capability provides a level of multi-dimensionality that existing literature might have overlooked. Furthermore, PLS-SEM is widely used in social science and tech adoption studies, but remains underutilised in material sciences in the aerospace and automotive industry [16].

This study makes three main contributions. First, it develops a model that shows how design, manufacturing, operational, and economic factors work together in the application of composite materials. Second, it gives practical and quantitative guidance to engineers and managers by highlighting more effective applications of composites in real-life projects. Finally, it puts forth some new evidence on how composites can move from the lab to large-scale use, looking at challenges such as cost, durability and scalability which must be conquered to achieve a more widespread utilisation of composites.

The rest of this paper follows this structure – Section 2 reviews applicable extant literature on the applications of composites in the aerospace and automotive industries. Section 3 details the research methodology, including the use of PLS-SEM and research hypotheses. The subsequent sections detail the empirical results (Section 4), discuss the findings in the aerospace and automotive contexts (Section 5), and conclude with key implications, study limitations, and directions for further research (Section 6)

II. LITERATURE REVIEW

This section examines prior studies on composite implementation across automotive and aerospace domains, emphasizing those that address the history, performance, structure, types, manufacturing processes, sustainability, and future scope.

First, the papers regarding the history and evolution of composites were analysed, finding that composites have slowly but surely entered mainstream engineering as a result of continuous progress and innovation [17]. For example, in the automotive industry, CMs were first utilised for racing or luxury vehicles, as performance and comfort respectively were prioritised (with a high cost ceiling). As the production and manufacturing of CMs become more efficient and widespread, they moved towards more mainstream and commercial applications [12]. However, the complexity of composite manufacturing and the limitations of conventional manufacturing meant that the full potential of composites was

not reached. Thus, techniques such as Additive Manufacturing (AM) were born, which used a unique layer-by-layer fabrication technique to produce complex polymer composites [18]. From 2014 to 2024, a bibliometric analysis found that AM grew substantially at a rate of 20.65% annually, underlined by significant increases in publications[19]. This is shown clearly in Figure 3, showing exponential year-on-year growth in AM-based research.

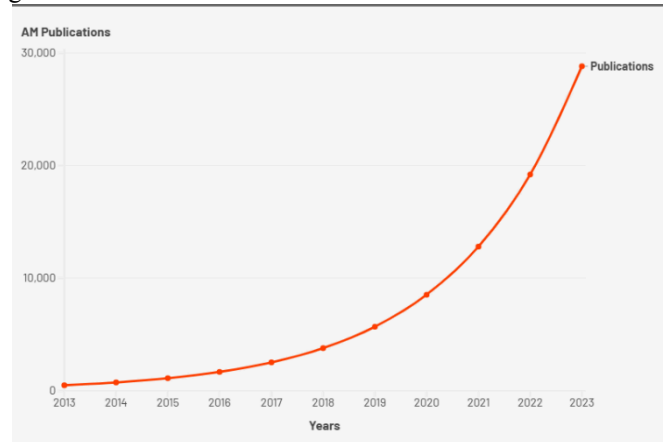


Fig. 3: Exponential growth in AM-related research

For AM to be successful and well-utilised, DfAM (Design for Additive Manufacturing) had to be optimized as well. DfAM is a set of guidelines followed by product designers to achieve high levels of success in printed designs. Consequently, improvements in DfAM brought unprecedented improvements in the properties and performance of some composites [20]. Subsequently, the literature surrounding material systems and material architectures – parameters that define the performance of composite materials – was analysed. CMs can be manufactured in a variety of different structures (namely, laminate stacking, woven/non-woven forms and hybrid structures), which are chosen based on their application and what it requires. To fulfil these requirements, different composites can be used, ranging from polymer matrix composites (PMCs), such as carbon fibre-reinforced plastics (CFRPs) and glass fibre-reinforced plastics (GFRPs), to metal matrix composites (MMCs), ceramic matrix composites (CMCs), and emerging bio-composites derived from natural materials [21]. Key parameters such as fibre orientation and microstructural characteristics were found to significantly impact properties like mechanical performance, strength, durability, and wear resistance [21]. In the two industries, both of which are based in human mobility, the architecture-dependent properties are crucial for ensuring structural integrity and crashworthiness [22]. Beyond traditional PMCs, there has been an exploration of the addition of some unconventional materials like marble dust and silica, with the intention of improving the mechanical properties of CMs further [22]

Research has found that a significant challenge in the CM

domain is assessing and modelling the long-term durability of composites: traditional modelling methods are costly and often take lot of time. To overcome this, the industry has moved towards multi-scale modelling, a form of computer-based analysis which predicts the durability and long-term behaviour of composites under different conditions [23]. These modelling techniques provide detailed information of degradation phenomena, such as hydrolytic, thermo-oxidative and chemical degradation processes [23].

Following this, it is important to transition to literature regarding manufacturing processes and technologies. Without these techniques, the superior properties and advantages of composites would be rendered ineffective. These technologies include CAM (Composite Additive Manufacturing, as mentioned before), quick-cure thermosets and thermoplastics, smart tooling and novel joining techniques [18]. CAM, specifically, is highly effective as it has allowed for a functional integration of composites into larger structures in aerospace. However, despite the significant advantages, research has shown that a more widespread adoption of CAM is hindered by a lack of certification, a lack of deep understanding of materials used, and the lack of a robust quality control system [18]. This shows the need for a more holistic approach which considers the performance of the system as a whole rather than individual performance metrics. Another important yet often overlooked aspect of composite manufacturing and application is quality control and testing. Advanced Non-Destructive Testing (NDT) approaches, including ultrasonic scanning, radiographic imaging, and acoustic emission monitoring, are used to find invisible cracks and damages that can seriously impact the structural integrity of structures [24]. NDT techniques can be applied to all types of composites. Here, the main challenge is the constant need for innovation and modernisation of NDT techniques to keep up with the growing complexity of composites [24].

Having established the fundamental material systems and manufacturing processes, the next step is an analysis of the literature surrounding the applications of composites in the aerospace and automobile industries. In aerospace, the applications are extensive, ranging from primary structures such as wings and fuselages (in both commercial aeroplanes like the Boeing 787 Dreamliner and military aircraft like the Lockheed Martin F-35 fighter), to other functions in propulsion systems and as functional coatings [25]. In automobiles, composites are used extensively in structural components like the chassis. Both applications require materials that have high load bearing, reliability, and comply with stringent regulations. Specific CMs, such as High-molecular reinforced Carbon Fibre systems (CFRH) are selected in aerospace contexts due to optimized matrix-reinforcement interactions suited to load requirements [26]. Performance, however, is not only limited to the matrix and reinforcement phases: further performance enhancements, namely nanofillers such as NanoSiC particles, are highly effective. Research has showed that implementing

nanofillers (such as coreshell rubber (CSR) or carboxyl-terminated butadiene acrylonitrile (CTBN)) can lead to improvements in fracture toughness and interlaminar properties. One study indicated a 250% increase in Mode I fracture toughness of a Carbon Fibre Reinforced Polymer (CFRP) after nanosized rubber particles were dispersed in the structure [27]. While these techniques possess an advantage in experimental and laboratory settings, high costs, complex manufacturing, certification problems, a lack of sustainability, and recyclability are some barriers to adoption [26][27].

Although they have become indispensable in the aerospace and automobile industries, there remain some barriers that limit a more widespread adoption of composites (as identified by existing literature). Technically, durability prediction in real-time environments remains uncertain, and emerging systems like self-healing composites are still in experimental phases. In terms of manufacturing, processes like CAM are highly promising but limited by a lack of robust quality control, certification issues and the lack of a holistic approach. Similarly, high production costs, immature supply chains and a lack of sustainability limit the cost-competitiveness of composites. Lastly, certain methodological gaps persist as well - most studies, as mentioned before, analyse metrics in isolation rather than in unison; techniques such as PLS-SEM remain underused in this sector.

III. RESEARCH METHODOLOGY

A. Research Objectives

This study investigates the structural and economic determinants governing composite material adoption in aerospace and automotive industries, with particular emphasis on the interdependence of these variables. We used the PLS-SEM (Partial Least Squares Structural Equation Modelling) framework as it allowed us to analyse the relationships between several key factors such as design, manufacturing processes, performance and cost savings. The goal is to understand how these factors collectively influence and impact cost-effectiveness, large-scale adoption and the long-term durability of composites in these two industries. Ultimately, the research hopes to move beyond the isolated study of these factors and assist in creating some form of guidelines for engineers, researchers and other stakeholders in the industry.

B. Research Questions

1. How do design and manufacturing parameters of composite materials influence operational performance in aerospace and automotive applications?
2. What role do cost-effectiveness and economic factors play in determining the scalability and adoption of composite materials in real-world projects?

- How do durability and performance outcomes collectively affect the long-term industrial utilisation of composites?

C. Hypotheses

- Design and manufacturing parameters have a significant positive impact on the operational performance of composite materials.
- Cost-effectiveness and economic feasibility significantly influence the large-scale adoption of composite materials.
- Durability mediates the relationship between operational performance and the long-term utilisation of composite materials.

D. Research Framework

Studying the adoption of composites involves multiple interacting factors that cannot be studied independently. Some of these factors are qualitative and difficult to measure as well, creating the need for latent constructs. A modelling approach is needed that considers the interrelationships of different variables simultaneously and also supports exploratory analysis. The PLS-SEM technique, a multivariate analysis technique enabling a simultaneous and concurrent analysis of multiple variables and supporting a relatively small sample size, fulfilled these requirements. The framework is structured around 3 latent constructs, namely Design and Manufacturing (R1), Cost-effectiveness (R2), and Durability/Long-Term Utilization (R3). Each of these constructs addresses a different factor influencing composite adoption. The survey items cover a variety of sub-questions under the domains of the respective constructs. Primary data were gathered through a systematically constructed online survey distributed to professionals, managers, and researchers across aerospace and automotive sectors

- Sample size:** 74 respondents
- Sampling technique:** purposive sampling
- Data type:** primary, collected via online forms
- Measurement scale:** 5-point Likert scale (1 = strongly disagree, 5 = strongly agree)

The questionnaire comprised 15 Likert-scale items, with five items corresponding to each latent construct. The three constructs were derived from the research questions (as mentioned before). The questions were developed with expert validation and thorough analysis of extant literature.

Table I: Survey Items with Outer Loadings

Construct	Survey Items	Example	Outer Loadings
Design & Manufacturing (DM)	Q1- Q5	“Advanced manufacturing techniques positively	0.820, 0.747, 0.813,

		affect the efficiency of composites in real-world projects”	0.892, 0.507
Cost effectiveness (CE)	Q6 - Q10	“Reducing the cost of composites is critical for their wider adoption in the automotive and aerospace industries”	0.687, 0.789, 0.773, 0.746, 0.767
Durability and Utilization (DU)	Q11- Q15	“The durability of composites is a deciding factor in their long-term adoption”	0.695, 0.789, 0.868, 0.864, 0.875

Table 1 shows the alignment of the survey items (questions) to the constructs, quantified by the outer loadings from the Smart-PLS 4.0 algorithm. All items demonstrate satisfactory reliability above the accepted thresholds (0.7), with only two items slightly below the thresholds. The data was pre-processed in order to remove any incomplete or incoherent responses. The normality of the data was ensured via skewness and Kurtosis checks.

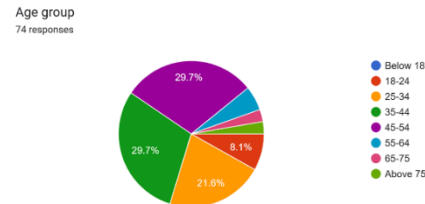


Fig. 5: Age distribution of respondents

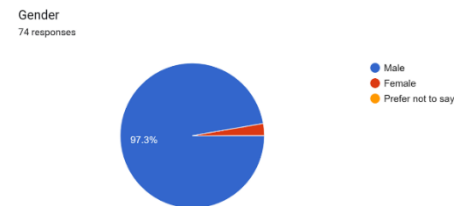


Fig. 6: Gender of respondents

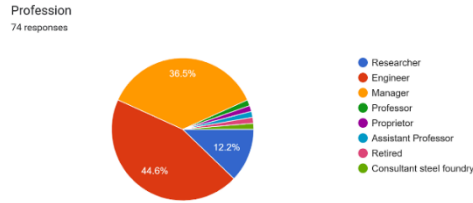


Fig. 7: Profession of respondents

Figures 5-7 show the demographics of respondents to the survey form.

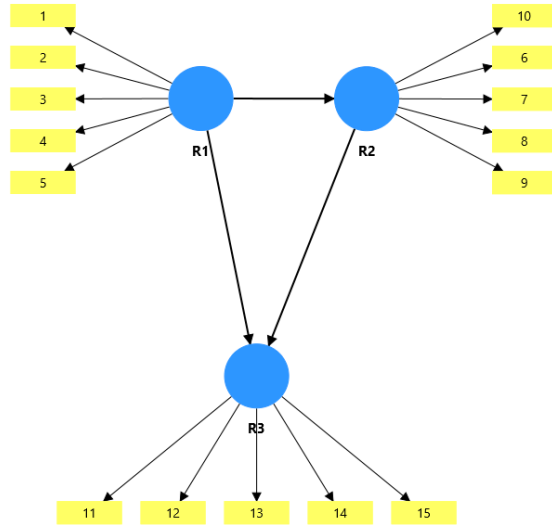


Fig. 8: Mapping research questions to survey items

Figure 8 is a visual representation of the relationships between the constructs and the questions for ease of understanding, taken from the Smart-PLS software. The methodology flow shown in Figure 9 is directly aligned with each of the study's three latent constructs. The PLS-SEM approach enables a quantifiable data-backed understanding of how the constructs influence the adoption of CMs in the two industries. It also allows for statistical robustness, yielding results and data that could be beneficial for material scientists and engineers looking to optimise the use of composites in these fields.

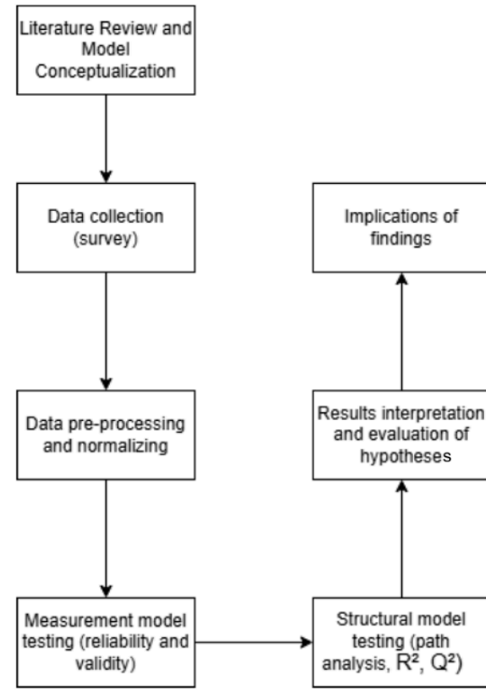


Fig. 9: Flow outlining research methodology

IV. RESULTS AND DISCUSSION

The extant literature attributes the superior performance of CMs mainly to manufacturing techniques such as curing precision, fibre placement, and laminate architecture, as shown in NASA's 2016 report on the "Proceedings of a NASA-sponsored Technical Interchange Conference" [6]. Separately, other studies delve into the economic challenges surrounding the adoption of composite materials [8], demonstrating how supply-chain issues, delays in certification and high tooling costs prevent a larger-scale adoption. However, the two studies predominantly consider these parameters in distinct "silos", an approach that is similar to that of most existing research regarding CMs in this aspect. This study integrates these factors using the PLS-SEM model, creating a robust and reliable framework to properly analyse the determinants of CM adoption.

A. Reliability and Validity Assessments

Before moving on to the analysis of the structural model, it was crucial to determine whether the model had strong reliability and validity. Subsequent assessments found the model to have strong internal consistency and minimal redundancy or overlap between constructs, showing strong distinctness. Table 2 provides the results for specific reliability and validity assessments.

Table 2: Survey Items with Outer Loadings

Parameter	Values	Interpretation
Cronbach's α	0.816-0.877	Strong internal consistency
Composite Reliability	>0.86	Exceeding the 0.7 threshold
Average Variance Extracted (AVE)	>0.5	Exceeding the 0.5 threshold
Fornell Larcker Criterion	Square roots of AVE values > inter-construct correlations	Satisfying condition
HTMT Ratios	0.788-0.888	Below the 0.9 threshold

B. Structural Model Analysis

The PLS-SEM analysis revealed significant dependencies between the constructs:

- Design & Manufacturing $\rightarrow$ Cost-Effectiveness ($\beta = 0.734$, $p < 0.01$)
- Design & Manufacturing $\rightarrow$ Durability/Utilisation ($\beta = 0.688$, $p < 0.01$)
- Cost-Effectiveness $\rightarrow$ Durability/Utilisation ($\beta = 0.670$, $p < 0.05$)

The values indicate a domino effect between the constructs, showing that improvements in Design and Manufacturing would lead to increases in Cost-Effectiveness. Similarly, increases in cost-effectiveness were found to have an impact on Durability and Utilisation. The most significant relationship was determined to be the one between Design and Manufacturing and Cost-Effectiveness. R^2 values of 0.538 and 0.532 for cost-effectiveness and durability, respectively, indicate a moderate to strong explanatory power.

C. Interpretation and Insights

While prior literature, such as NASA's 2016 Report[6] and the ATI Insight Report [9], concentrates on individual factors such as design and manufacturing and economic barriers, respectively, this study represents a more holistic approach. By integrating these factors with PLS-SEM, it validates how manufacturing quality, both directly (through strength and structural integrity) and indirectly (through cost-savings),

exerts a significant and statistically proven direct and indirect influence on composite adoption within the automobile and aerospace sectors. This dual impact shows how Design and Manufacturing have a compounded effect on the adoption of CMs. This relationship is shown visually in Figure 10.

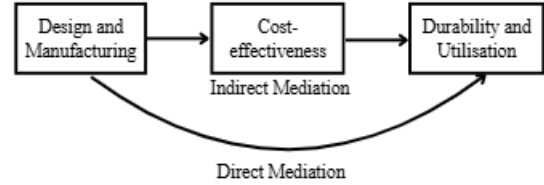


Fig. 10. Proposed framework and flowchart

D. Policy and Industrial Implications

1. Establish standardisation protocols for composites, ensuring interchangeability across industries and countries, potentially reducing design redundancy by 18–20%.
2. Promote structured R&D collaboration in the aerospace and automotive sectors to achieve around 25% cost savings in material innovations.
3. Integrate access to AI-driven structural simulation software tools during the design process to minimise prototype costs and wastage by 30%.
4. Introduce sustainability-linked credits for recyclable, long-lived composites to encourage adoption by firms.

There are some possible improvements and limitations to this study. With a relatively small number of 74 respondents, the study's conclusions should be generalized with caution. Focusing solely on the automobile and aerospace industries means the study may not reflect trends or practices in other composite-heavy industries like marine or architectural engineering. Additionally, because the study uses cross-sectional data, it cannot capture how views on composite adoption may shift over time. Future research in this area may benefit from larger, longitudinal datasets and expanding the scope to include more than 2 industries. Lastly, accounting for environmental or technological innovation with additional constructs may make the research more applicable.

V. CONCLUSION

In summary, our findings suggest that the adoption of composite materials across the automobile and aerospace sectors is dependent on the combined influence of multiple factors, namely design and manufacturing, cost-effectiveness, and durability and utilization. The findings demonstrate that design and manufacturing have a dual and compound impact on the adoption of CMs, designating it as a crucial factor. Still, the successful implementation of composites depends on a holistic understanding and application of all the factors rather than any single factor. These data-backed relationships offer practical benefits for aerospace and automotive manufacturing

firms, reinforcing predictive decision models applicable across engineering domains. The policy insights underscore the transition towards optimized, data-informed manufacturing ecosystems. By addressing the research gap through an interconnected analysis, this study provides a foundation for sustainability-driven innovations surrounding composites in the future.

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